E-commerce Sales Data Analysis

PROJECT OBJECTIVE:

The Objective of this Capstone Project is to Analysis E-commerce Sales data Using Microsoft Excel, To Perform Data cleaning and transformation, Apply Descriptive Analysis techniques and generate Actionable Insights through Visualizations and create Interactive Dashboards to Support decision-making.

DATASET DESCRIPTION:

The dataset Contains Multiple sheet, Includes, Sales_fact, Customer_dim, Product_dim,Store_dim Tables. It contains Customer details, Product details, Store details, Sales details.

It Contains transactions sales records, product details, pricing. Discounts, quantity sold, stock information and customer purchase history used or analytical reporting.

COLUMN DESCRIPTION:

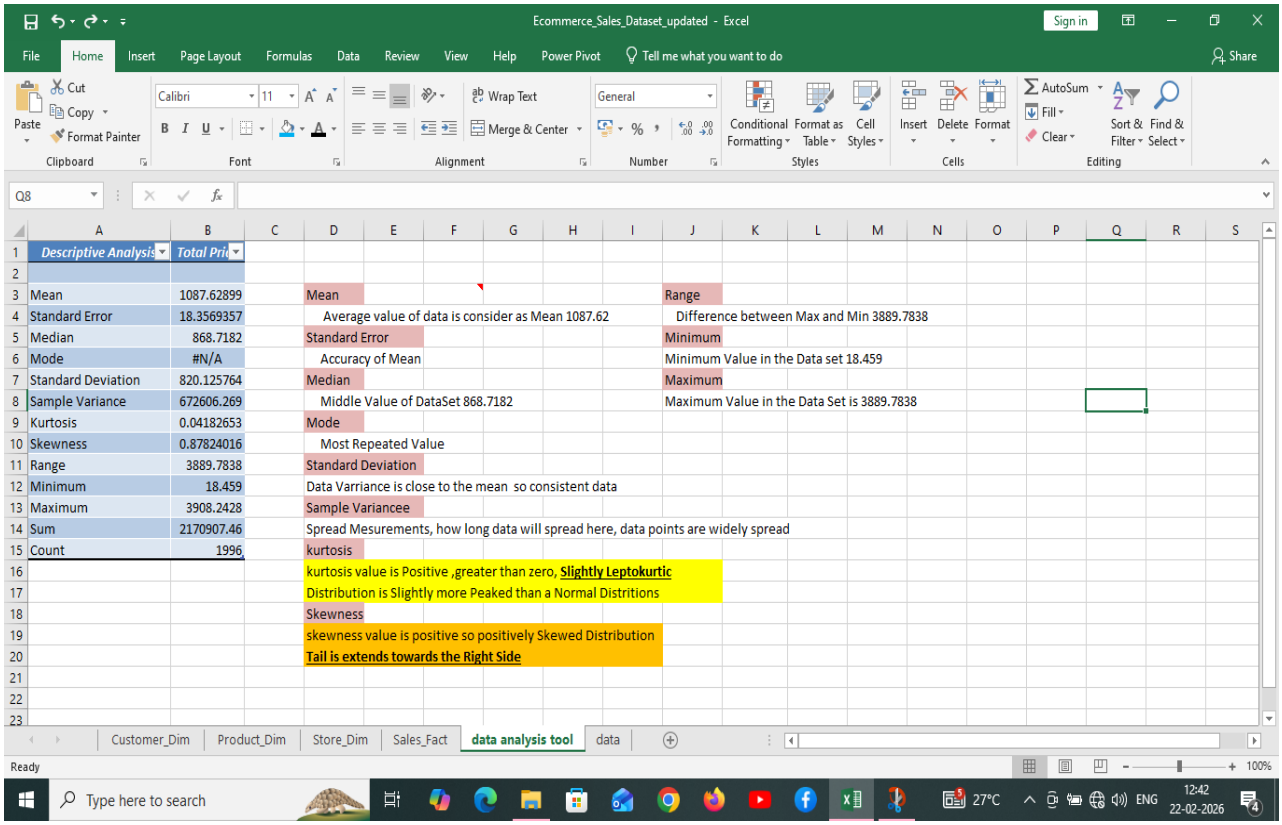
COLUMN NAME	DESCRIPTION
Customer Id	Unique Identifier for each customer
Product Id	Unique Identifier for each Product
Sales Id	Unique Identifier for each Sale
Store Id	Unique Identifier for each store
Loyality	Rewards Points to the customer
category	Product Category
Stock	Available stock in each product
Cost	Cost of each Product
Quantity	Quantity of each product
Unit Price	Unit Price of each product
Discounts	Discounts of each product

Total Price	Total Price Of Products
Date	Order date of Products
Region	Region of each Store

Functions used in Projects

FUNCTIONS	Column	PURPOSE
=TRIM(CLEAN(Table1[[#All],[Product_Name]]))	Product table	To **clean** and **trim** the data
=COUNTIF (H2:H501,"Platinum")	Customer Loyality	To find out mode using **countif**
IF(ISBLANK(I5:I505),"Platinum",I5)	Customer Loyality	To fill blank using **isblank**
Find and Replace	Product Id	To change product id
=IF(C3>60,"Senior Citizen",IF(C3>40,"Middle age ",IF(C3>30,"Adults Age",IF(C3>20,"Young People","Teenage"))))	Age Catagory	Using **Nested if** function create custom column
=PROPER(B2:B501)	Customer Name	**Proper** case for customer name
=IF(ISBLANK(F2),AVERAGE(F2:F 102),F2)	Product cost	Find average cost and fill null values using **if(isblank** function
IF(ISBLANK(G3),AVERAGE(G3:G2002),G3)	Quantity in sales table	Find average of Quantity and fill null values using **if(isblank** function
VLOOKUP(E3,Product_Dim!A:B,2,FALSE)	PRODUCT NAME	Using customerid in sales table retrive custome name using **VLOOKUP**
=[@Quantity]*[@[Unit_Price]]*(1-[@Discount])	Total Price in Sales table	To find out total Price
=SUM(J2:J2001)	Total Price in Sales table	To find total price using **SUM** function

Data Analysis Tool Pak



Descriptive Analysis	Total Pri
Mean	1087.62899
Standard Error	18.3569357
Median	868.7182
Mode	#N/A
Standard Deviation	820.125764
Sample Variance	672606.269
Kurtosis	0.04182653
Skewness	0.87824016
Range	3889.7838
Minimum	18.459
Maximum	3908.2428
Sum	2170907.46
Count	1996

Mean
Average value of data is consider as Mean 1087.62
Standard Error
Accuracy of Mean
Median
Middle Value of DataSet 868.7182
Mode
Most Repeated Value
Standard Deviation
Data Varriance is close to the mean so consistent data
Sample Variancee
Spread Mesurements, how long data will spread here, data points are widely spread
kurtosis
kurtosis value is Positive ,greater than zero, Slightly Leptokurtic
Distribution is Slightly more Peaked than a Normal Distritions
Skewness
skewness value is positive so positively Skewed Distribution
Tail is extends towards the Right Side

Range
Difference between Max and Min 3889.7838
Minimum
Minimum Value in the Data set 18.459
Maximum
Maximum Value in the Data Set is 3889.7838

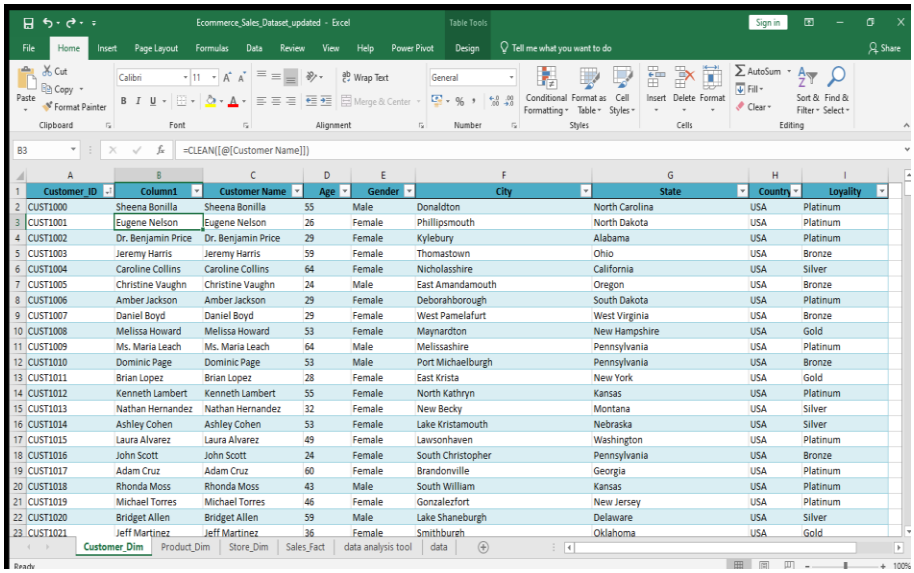
Important points:

kurtosis

- kurtosis value is Positive, greater than zero, Slightly Leptokurtic
- Distribution is Slightly more Peaked than a Normal Distritions

Skewness

- skewness value is positive so positively Skewed Distribution
- Tail is extending towards the Right Side



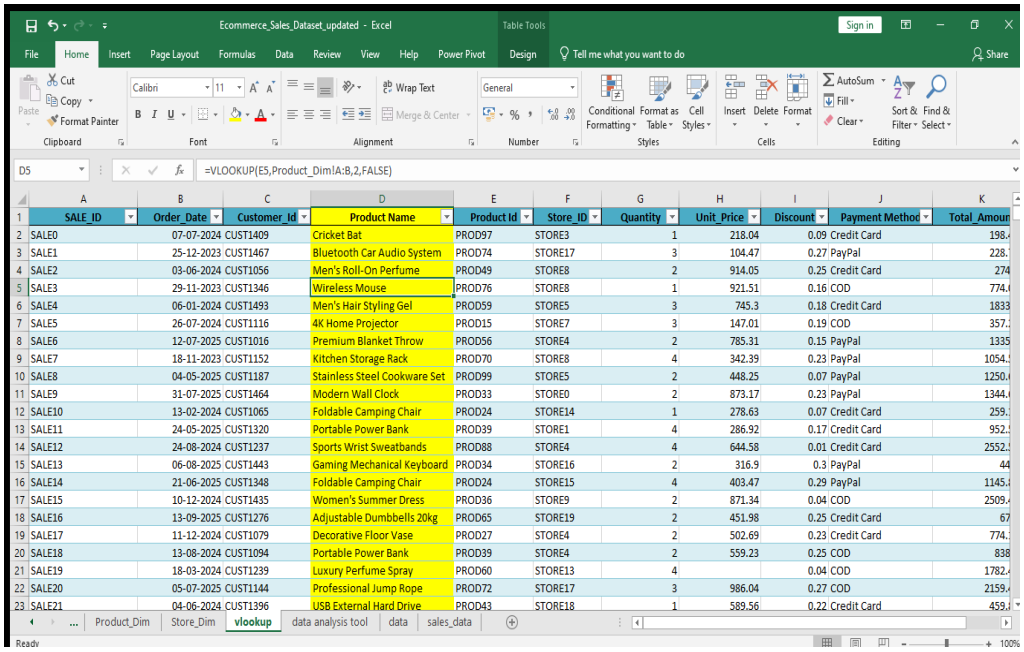
Ecommerce_Sales_Dataset_updated - Excel

B3 =CLEAN([@[Customer Name]])

Customer_ID	Column1	Customer Name	Age	Gender	City	State	Country	Loyality
CUST1000	Sheena Bonilla	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Platinum
CUST1001	Eugene Nelson	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	Rhonda Moss	43	Male	South William	Kansas	USA	Platinum
CUST1019	Michael Torres	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold

Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | data analysis tool | data

VLOOKUP FUNCTION

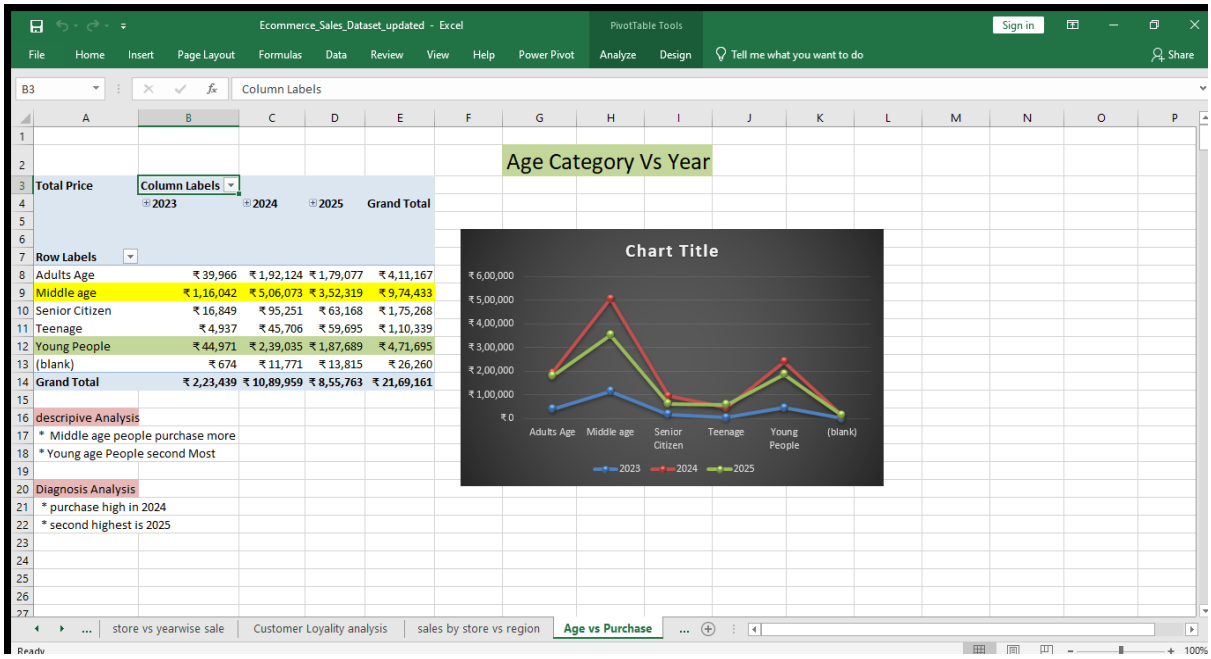


Ecommerce_Sales_Dataset_updated - Excel

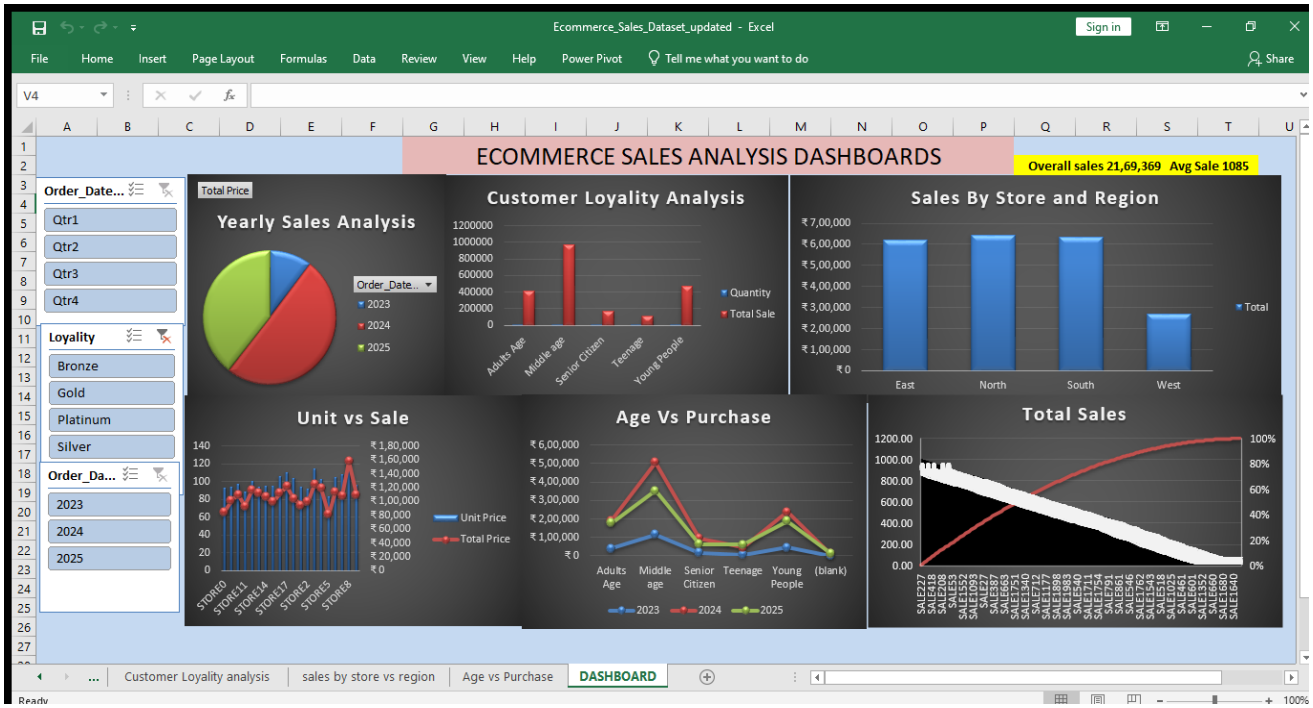
D5 =VLOOKUP(E5,Product_Dim!A:B,2,FALSE)

SALE_ID	Order_Date	Customer_Id	Product Name	Product Id	Store_ID	Quantity	Unit_Price	Discount	Payment Method	Total_Amoun
SALE0	07-07-2024	CUST1409	Cricket Bat	PROD97	STORE3	1	218.04	0.09	Credit Card	198.
SALE1	25-12-2023	CUST1467	Bluetooth Car Audio System	PROD74	STORE17	3	104.47	0.27	PayPal	228.
SALE2	03-06-2024	CUST1056	Men's Roll-On Perfume	PROD49	STORE8	2	914.05	0.25	Credit Card	274
SALE3	29-11-2023	CUST1346	Wireless Mouse	PROD76	STORE8	1	921.51	0.16	COD	774.
SALE4	06-01-2024	CUST1493	Men's Hair Styling Gel	PROD59	STORE5	3	745.3	0.18	Credit Card	1833
SALE5	26-07-2024	CUST1116	4K Home Projector	PROD15	STORE7	3	147.01	0.19	COD	357.
SALE6	12-07-2025	CUST1016	Premium Blanket Throw	PROD56	STORE4	2	785.31	0.15	PayPal	1335
SALE7	18-11-2023	CUST1152	Kitchen Storage Rack	PROD70	STORE8	4	342.39	0.23	PayPal	1054.
SALE8	04-05-2025	CUST1187	Stainless Steel Cookware Set	PROD99	STORE5	2	448.25	0.07	PayPal	1250.
SALE9	31-07-2025	CUST1464	Modern Wall Clock	PROD33	STORE0	2	873.17	0.23	PayPal	1344.
SALE10	13-02-2024	CUST1065	Foldable Camping Chair	PROD24	STORE14	1	278.63	0.07	Credit Card	259.
SALE11	24-05-2025	CUST1320	Portable Power Bank	PROD39	STORE1	4	286.92	0.17	Credit Card	952.
SALE12	24-08-2024	CUST1237	Sports Wrist Sweatbands	PROD88	STORE4	4	644.58	0.01	Credit Card	2552.
SALE13	06-08-2025	CUST1443	Gaming Mechanical Keyboard	PROD34	STORE16	2	316.9	0.3	PayPal	44
SALE14	21-06-2025	CUST1348	Foldable Camping Chair	PROD24	STORE15	4	403.47	0.29	PayPal	1145.
SALE15	10-12-2024	CUST1435	Women's Summer Dress	PROD36	STORE9	2	871.34	0.04	COD	2509.
SALE16	13-09-2025	CUST1276	Adjustable Dumbbells 20kg	PROD65	STORE19	2	451.98	0.25	Credit Card	67
SALE17	11-12-2024	CUST1079	Decorative Floor Vase	PROD27	STORE4	2	502.69	0.23	Credit Card	774.
SALE18	13-08-2024	CUST1094	Portable Power Bank	PROD39	STORE4	2	559.23	0.25	COD	838
SALE19	18-03-2024	CUST1239	Luxury Perfume Spray	PROD60	STORE13	4		0.04	COD	1782.
SALE20	05-07-2025	CUST1144	Professional Jump Rope	PROD72	STORE17	3	986.04	0.27	COD	2159.
SALE21	04-06-2024	CUST1396	USB External Hard Drive	PROD43	STORE18	1	589.56	0.22	Credit Card	459.

... | Product_Dim | Store_Dim | vlookup | data analysis tool | data | sales_data



Ecommerce Sales Analysis Dashboards



Four Types of Analysis

Descriptive Analysis:

The Sales performance was analyzed using available data for the years 2023,2024 and 2025.Complete annual data is available for 2024 and 2025. While only fourth-quarter data is available for 2023. **2024** has **highest sale**. **Online Store has high sale** comparative flagship and outlet. Two regions have outlet store North,south.

Middle age people purchase more than other age category people. In 2025 last Quarter of the has lowest Sales Unit Price is in high in 2024 that is the reason ,2024 has high sale.

A Comparative Analysis between 2024 and 2025 shows a **10% decline in total sales** in **2025.** The 2023 Q4 data indicates moderate sales performance but cannot be directly comapared annually due to incomplete records. Over all sales show a declined trend from 2024 to 2025

Diagnosis Analysis:

The decline in sales during 2025 may be attributed to several factors. Possible Contributing reasons include **Reduced customer demand**, **increased market competition**, **pricing adjustments** or Operational Challenges affecting Product availability.

Compared to 2024, lower Quarterly sale performance suggests decreased customer acquisition and retention. May be **2025 last Quarter sale is affected by year end period**.

Predictive Analysis:

Forecasting Future trends or behaviors sales like to increase by **15%** next Quarter due to **New Year 2026 Promotions and offers** and Festivals offers and give more discounts.

Prescriptive Analysis:

If **it possible to open outlet in West and east** Regions, it **will increase 20% sale in 2026.** Create Optimize **Website for Sales** will increase Sale more, **launch some molibe app** will also increasing sales in future year, because **Online sale is high.**

Conclusion:

The Analysis of sales data for 2023,2024 and 2025 provides valuable insights into business performance trends. Although a 10% decline in sale was observed from 2024 to 2025 and 2023 data was limited to the last quarter, the finding offers an opportunity to identify improvements areas and strengthen future strategies. With effective planning, enhance marketing efforts and data-driven decision making. The organization has strong potential to improve sales performance and achieve sustainable growth in the coming years.